# WaRee11

## iPadOS River Simulation & Hydraulic Modeling Platform

**Public White Paper Brief v0.1**


For partners, investors, university laboratories, academic collaborators, and technical advisors

Status: Concept / Prototype Direction | Date: May 2026

# 1. Executive Brief

WaRee11 is an iPadOS-based river simulation and hydraulic modeling platform designed to make river model preparation, hydraulic education, scenario exploration, and results communication more accessible through a touch-first iPad workflow.

The platform is inspired by professional hydraulic modeling systems such as HEC-RAS, MIKE 11 / MIKE+ Rivers, SOBEK, and D-Flow FM, but it is not positioned as a direct replacement at this stage. Its early role is best understood as a focused companion platform for learning, field preparation, model readiness checking, and structured scenario thinking.

WaRee11's initial direction emphasizes a one-dimensional river modeling workflow, cross-section management, simulation settings, CFL-aware diagnostics, scenario comparison, and future GIS-assisted model preparation. The long-term ambition is to connect classroom learning, field observation, and professional river modeling practice.

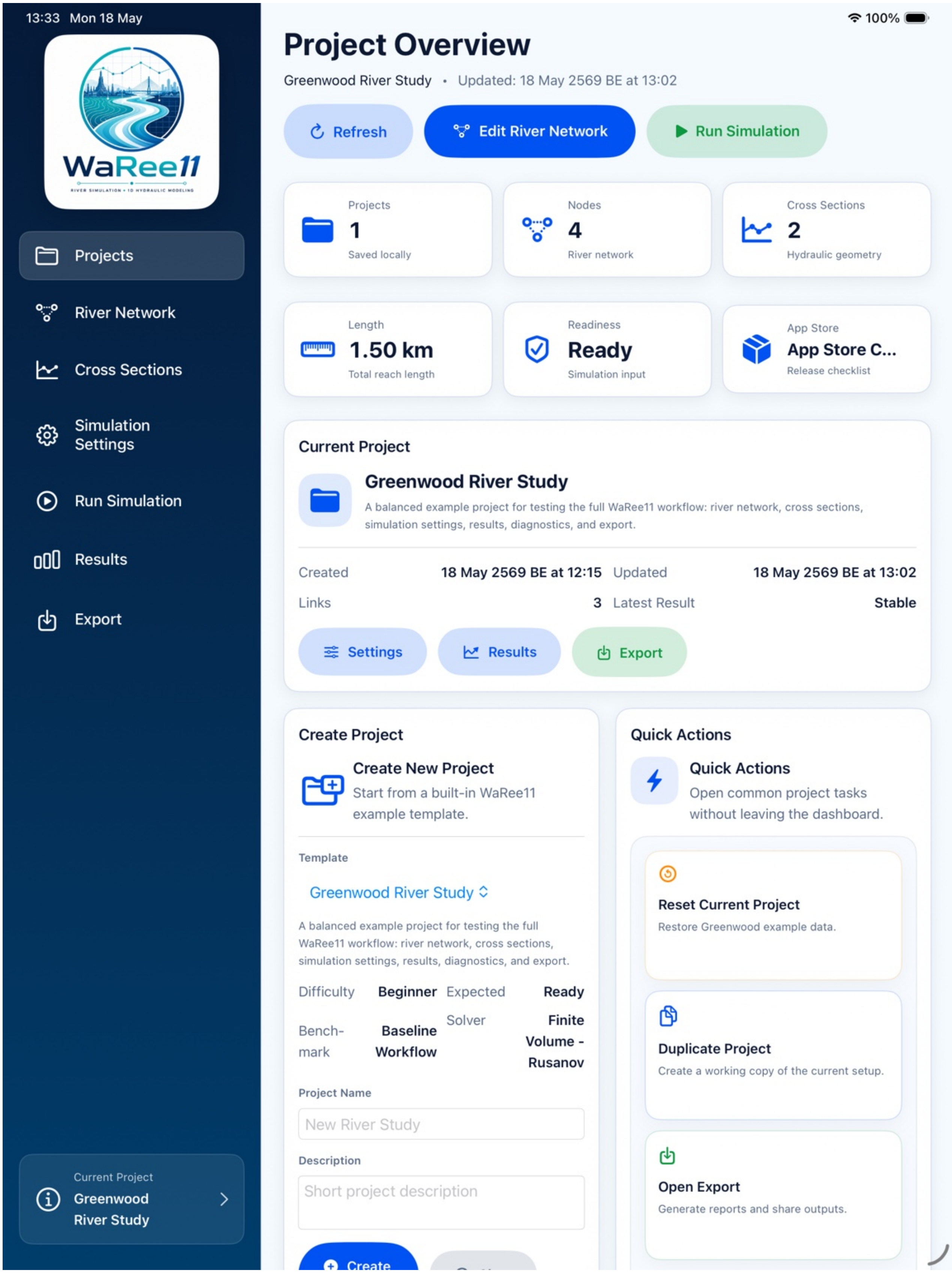


*Figure 1. WaRee11 project dashboard showing project readiness and workflow navigation.*

## 2. Why WaRee11 Matters

| Value Area | Contribution |
|---|---|
| **Education** | Provides students and instructors with a visual iPad environment for understanding river networks, cross-section geometry, Manning roughness, boundary conditions, CFL stability, and simulation results. |
| **Field preparation** | Helps engineers and field teams review river networks, cross-section completeness, roughness assumptions, and model readiness away from a desktop workstation. |
| **Scenario exploration** | Supports rapid what-if comparison of assumptions such as roughness, hydrographs, downstream water levels, and simplified design configurations. |

## 3. Positioning

WaRee11 should be presented as a touch-first iPadOS companion platform, not as a full replacement for mature professional desktop modeling systems. This positioning keeps the project credible while highlighting its unique role in education, field review, and early scenario analysis.

**Classroom Hydraulics → WaRee11 → Professional Desktop Modeling**

Supported by field GIS / iPad-based model preparation workflows

# 4. Core WaRee11 Modules

| Module | Role in the Workflow |
|---|---|
| **Project Dashboard** | Project health, readiness status, recent runs, and export shortcuts. |
| **River Network Editor** | River nodes, branches, chainage, connectivity, and topology checks. |
| **Cross-Section Editor** | Geometry input, Manning roughness, cross-section preview, hydraulic properties, and rating curve preview. |
| **Simulation Settings** | Time step, duration, CFL checking, numerical scheme selection, and pre-run diagnostics. |
| **Results & Diagnostics** | Water level charts, discharge charts, run summaries, solver warnings, result tables, and export-ready outputs. |

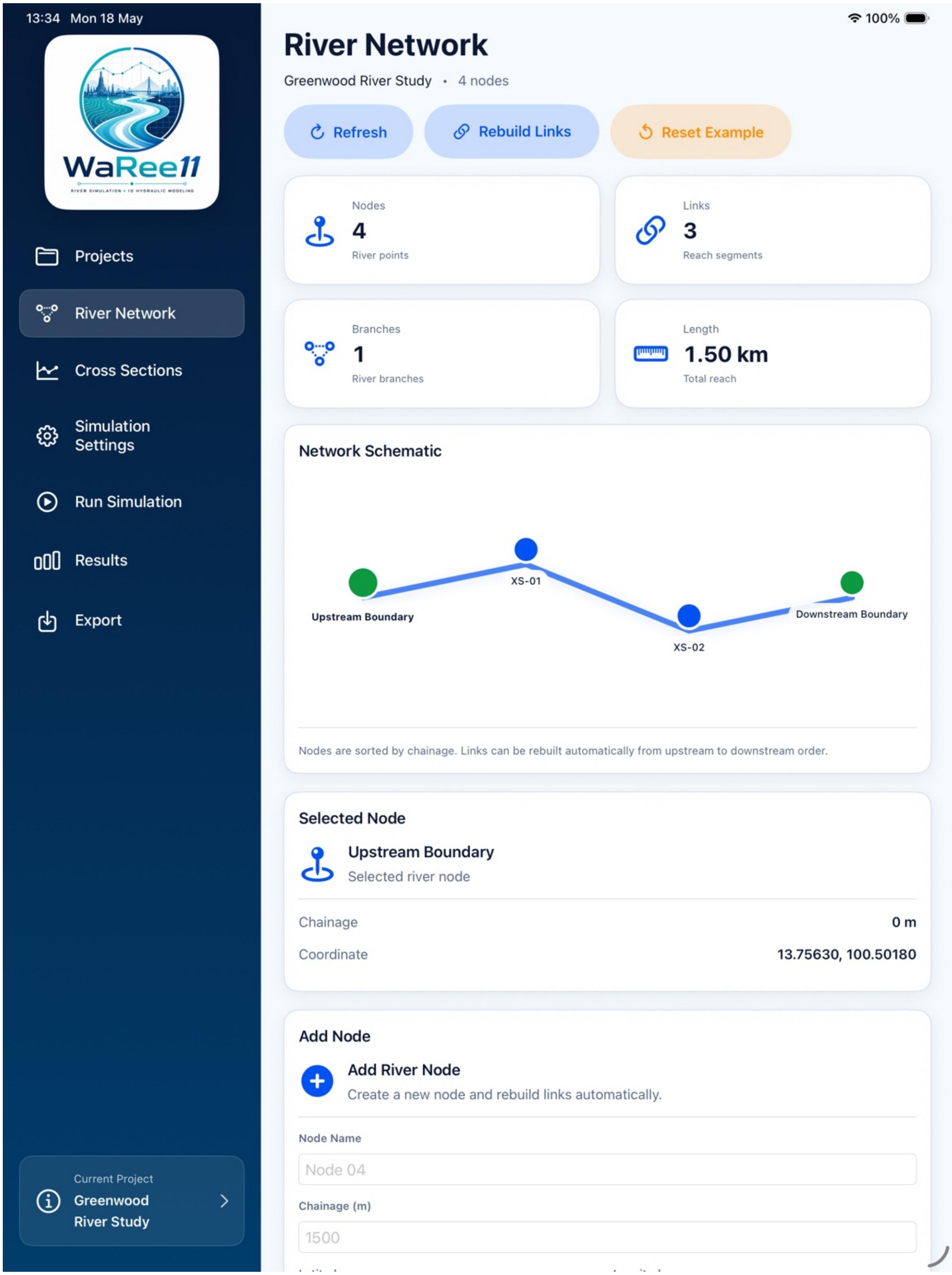


*Figure 2. River network editor showing nodes, links, branches, and reach length.*

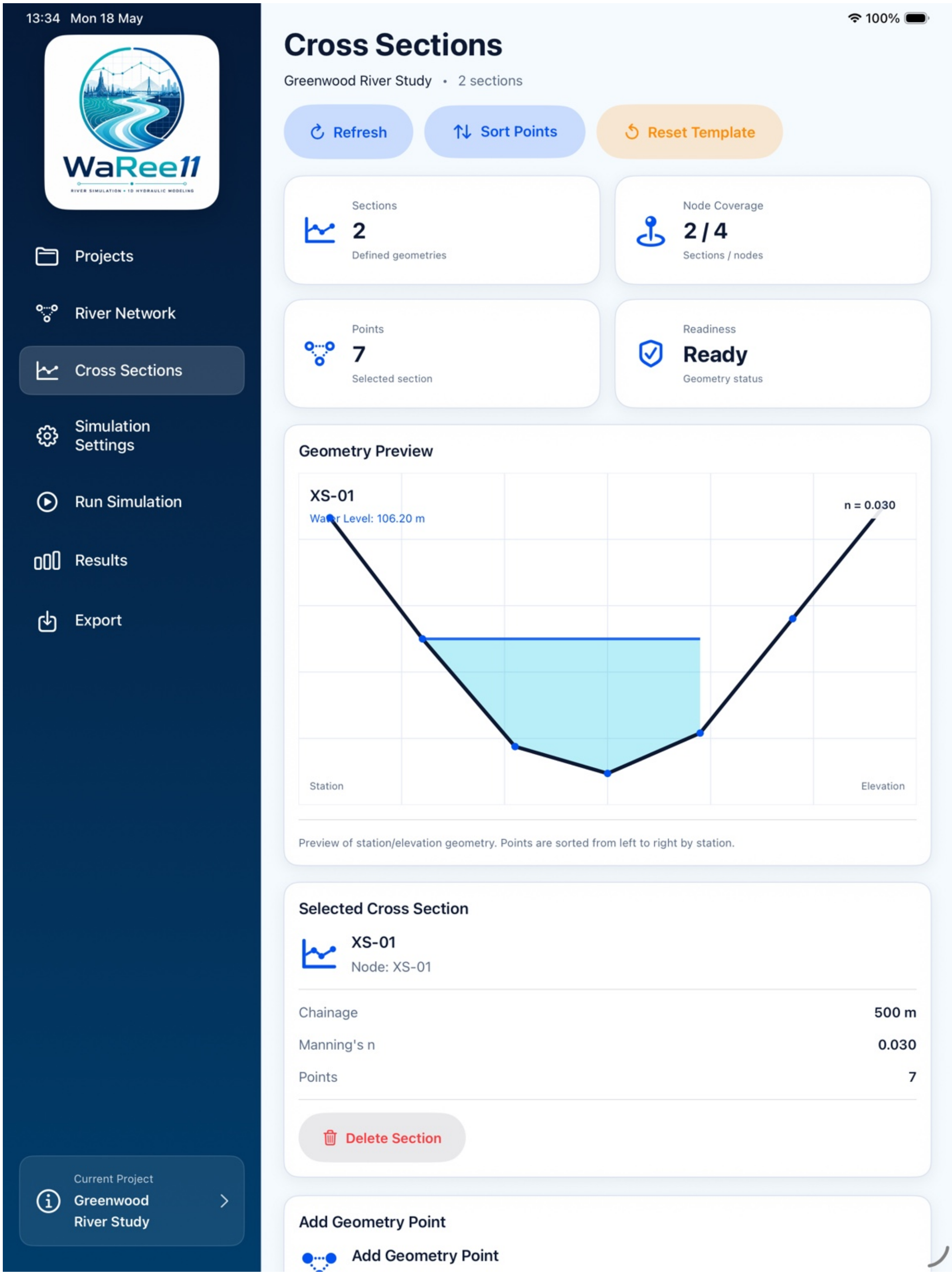


*Figure 3. Cross-section editor showing geometry preview and hydraulic section data.*

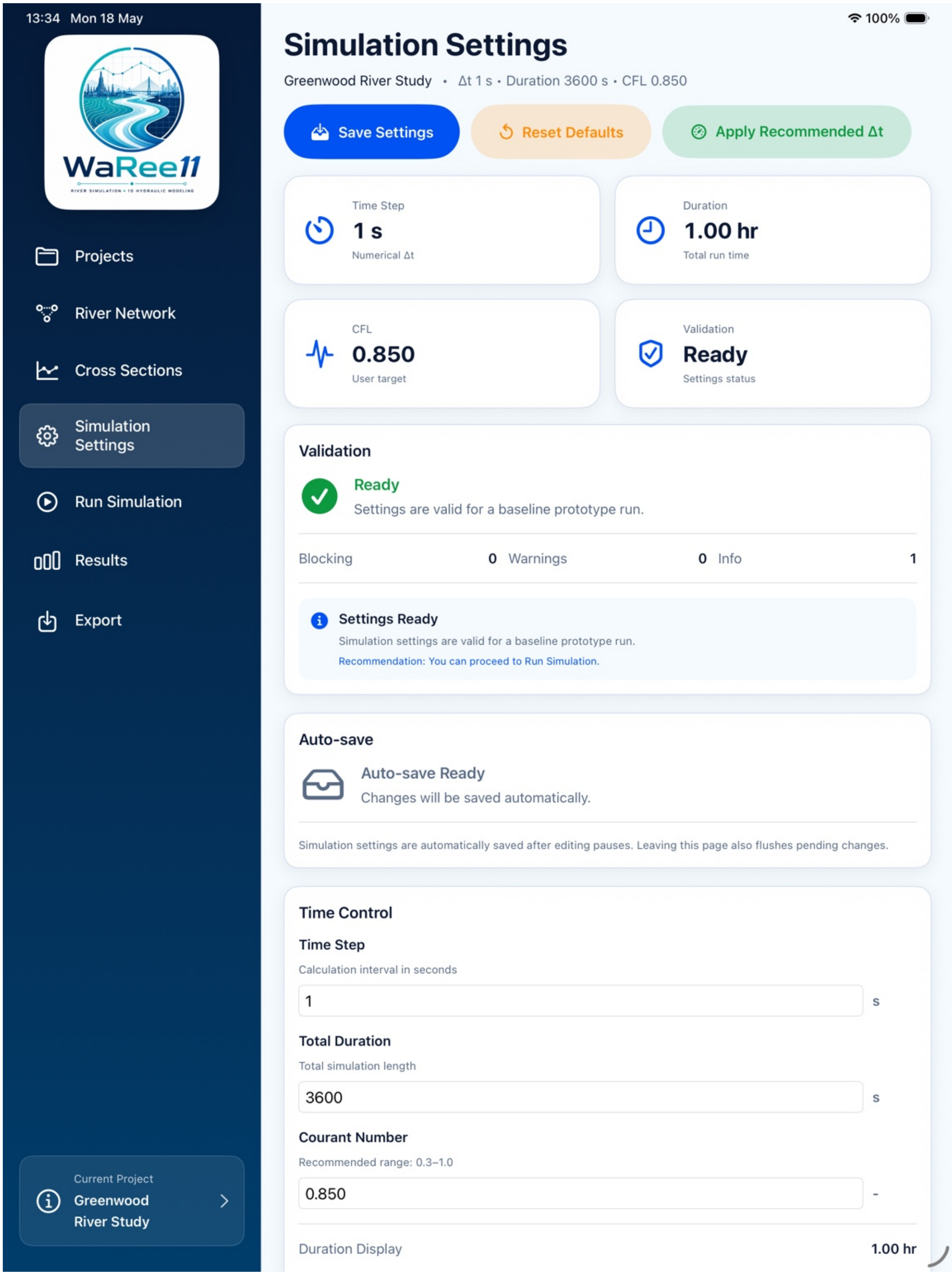


*Figure 4. Simulation settings with time step, duration, CFL target, and readiness validation.*

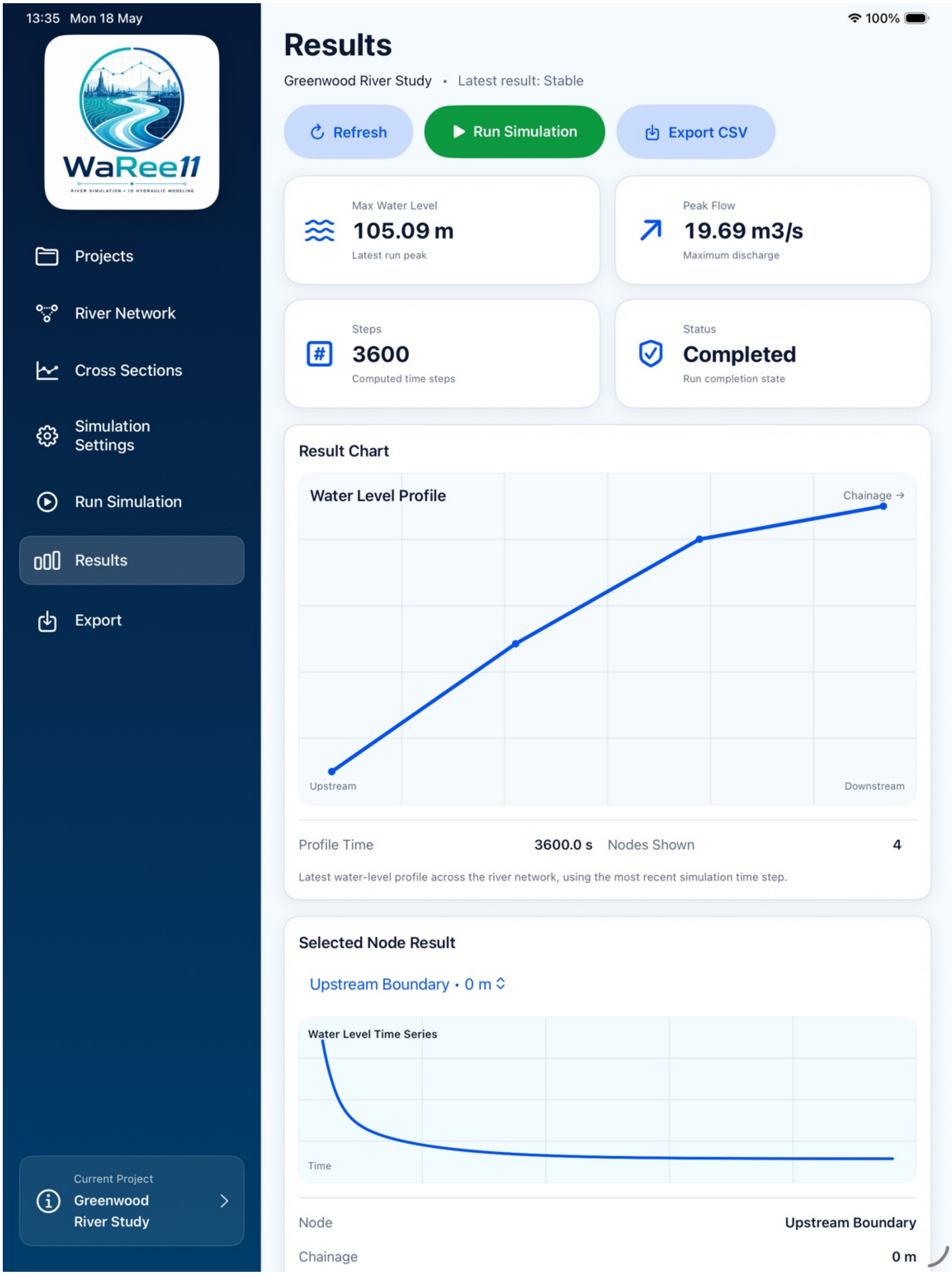


*Figure 5. Results screen showing water level profile, peak flow, run status, and output readiness.*

# 5. Technical Direction

WaRee11 is planned around a modern SwiftUI and MVVM architecture with repository-based data management, project-based storage, a modular solver engine, diagnostics, and export layers.

The hydraulic solver direction begins with a focused 1D river hydraulic workflow based on the Saint-Venant equation family. Development should progress through educational prototypes, explicit scheme comparisons such as MacCormack and Lax-Wendroff, finite-volume development, benchmark verification, and laboratory validation.

A central design principle is that numerical behavior should be visible. WaRee11 should help users understand time step selection, Courant number / CFL behavior, missing inputs, and whether a simulation result is suitable for interpretation.

## 6. GIS-Assisted Model Preparation

The future WaRee11 GIS editing layer should provide model-specific spatial editing for hydraulic schematization. The goal is not to build a full QGIS or ArcGIS replacement, but to support river nodes, branches, cross-section placement, roughness zones, attributes, GeoJSON/CSV interoperability, and hydraulic readiness checks inside the WaRee11 workflow.

## 7. Validation and Collaboration Roadmap

**Code Verification → Benchmark Testing → Inter-Model Comparison → Laboratory Validation → Field Validation**

WaRee11 is well suited to structured collaboration with university hydraulic laboratories, water engineering departments, research groups, irrigation and flood agencies, and professional engineering partners. A practical collaboration package could include a teaching pilot, a benchmark validation dataset, and a controlled hydraulic laboratory demonstration.

## 8. Development Roadmap

| Phase | Focus | Key Output |
|---|---|---|
| **Phase 1** | App Store Prototype | Core workflow, diagnostics, scenario management, and export basics. |
| **Phase 2** | Hydraulic Engine Strengthening | Solver input builder, CFL calculator, scheme comparison, benchmark cases. |
| **Phase 3** | GIS Editing Layer | Map canvas, layer manager, node/branch editing, cross-section placement. |
| **Phase 4** | Calibration and Lab Validation | Benchmark library, lab datasets, inter-model comparison, validation reporting. |
| **Phase 5** | Education and Professional Platform | Teaching modules, partner pilots, documentation, training datasets, professional reporting. |

## 9. Partner Ask

WaRee11 is seeking early collaboration and feedback in the following areas:

- Pilot course collaboration
- Benchmark validation datasets
- Hydraulic laboratory partners
- Technical feedback from river modeling experts

- Research collaboration on iPad-based hydraulic model preparation

**WaRee11 can become a bridge between classroom learning, field observation, hydraulic model preparation, and professional river modeling practice.**